Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-30.jpg":

$$C = \lim_{n\to\infty} \left[n^2 \left(1 - \sqrt[n]{\int_0^1 (1-x^n)^n \, dx} \right) - \ln(n) \right]$$

Detailed Solution

To evaluate the limit C, we first need to find a closed form or an asymptotic expansion for the integral inside the nth root. Let us denote the integral as I_n:

$$I_n = \int_0^1 (1-x^n)^n \, dx$$

Step 1: Evaluating the Integral

We can transform I_n using the substitution $u = x^n$. This implies $x = u^{1/n}$ and $dx = \frac{1}{n} u^{\frac{1}{n}-1} \, du$. The limits of integration remain 0 to 1.

$$I_n = \int_0^1 (1-u)^n \left(\frac{1}{n} u^{\frac{1}{n}-1} \right) du = \frac{1}{n} \int_0^1 u^{\frac{1}{n}-1} (1-u)^n \, du$$

This is the definition of the Beta function, $\mathrm{B}(z_1, z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1} \, dt$, where $z_1 = \frac{1}{n}$ and $z_2 = n+1$:

$$I_n = \frac{1}{n} \mathrm{B}\left(\frac{1}{n}, n+1 \right) = \frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right)\Gamma(n+1)}{\Gamma\left(n+1+\frac{1}{n}\right)}$$

Using the property $\Gamma(z+1) = z\Gamma(z)$, we have $\frac{1}{n}\Gamma\left(\frac{1}{n}\right) = \Gamma\left(1+\frac{1}{n}\right)$. Thus:

$$I_n = \frac{\Gamma\left(1+\frac{1}{n}\right) n!}{\Gamma\left(n+1+\frac{1}{n}\right)}$$

We can expand the denominator using the recursive property of the Gamma function:

$$\Gamma\left(n+1+\frac{1}{n}\right) = \left(n+\frac{1}{n}\right)\left(n-1+\frac{1}{n}\right)\cdots\left(1+\frac{1}{n}\right)\Gamma\left(1+\frac{1}{n}\right)$$

Substituting this back into I_n, the $\Gamma\left(1+\frac{1}{n}\right)$ terms cancel out:

$$I_n = \frac{n!}{\prod_{k=1}^n \left(k+\frac{1}{n}\right)} = \prod_{k=1}^n \frac{k}{k+\frac{1}{n}} = \prod_{k=1}^n \left(1+\frac{1}{nk}\right)^{-1}$$

Step 2: Asymptotic Expansion of $\ln(I_n)$

To analyze the nth root $\sqrt[n]{I_n} = (I_n)^{1/n}$, we take the natural logarithm:

$$\ln(I_n) = -\sum_{k=1}^{n} \ln\left(1 + \frac{1}{nk}\right)$$

Using the Taylor series expansion $\ln(1+x) = x - \frac{x^2}{2} + \mathcal{O}(x^3)$ for small x:

$$\ln(I_n) = -\sum_{k=1}^{n} \left(\frac{1}{nk} - \mathcal{O}\left(\frac{1}{n^2k^2}\right)\right) = -\frac{1}{n}\sum_{k=1}^{n} \frac{1}{k} + \mathcal{O}\left(\frac{1}{n^2}\right)$$

The sum $\sum_{k=1}^{n} \frac{1}{k}$ is the n-th Harmonic number H_n, which has the asymptotic expansion $H_n = \ln(n) + \gamma + \mathcal{O}\left(\frac{1}{n}\right)$, where γ is the Euler-Mascheroni constant. Substituting this into our expression:

$$\ln(I_n) = -\frac{1}{n}\left(\ln(n) + \gamma + \mathcal{O}\left(\frac{1}{n}\right)\right) + \mathcal{O}\left(\frac{1}{n^2}\right) = -\frac{\ln(n)}{n} - \frac{\gamma}{n} + \mathcal{O}\left(\frac{1}{n^2}\right)$$

Step 3: Expansion of the Root

We need the behavior of $\sqrt[n]{I_n} = \exp\left(\frac{1}{n}\ln(I_n)\right)$. Dividing our previous result by n:

$$\frac{1}{n}\ln(I_n) = -\frac{\ln(n)}{n^2} - \frac{\gamma}{n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)$$

Since this term approaches 0 as $n \to \infty$, we can use the Taylor series expansion $e^z = 1 + z + \mathcal{O}(z^2)$:

$$\sqrt[n]{I_n} = 1 + \left(-\frac{\ln(n)}{n^2} - \frac{\gamma}{n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)\right) + \mathcal{O}\left(\frac{\ln^2(n)}{n^4}\right)$$

Simplifying, we obtain:

$$\sqrt[n]{I_n} = 1 - \frac{\ln(n)}{n^2} - \frac{\gamma}{n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)$$

Step 4: Evaluating the Limit

Now, substitute this expansion back into the original limit expression for C:

$$C = \lim_{n\to\infty}\left[n^2\left(1 - \left(1 - \frac{\ln(n)}{n^2} - \frac{\gamma}{n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)\right)\right) - \ln(n)\right]$$

$$C = \lim_{n\to\infty}\left[n^2\left(\frac{\ln(n)}{n^2} + \frac{\gamma}{n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)\right) - \ln(n)\right]$$

Distribute the n^2:

$$C = \lim_{n\to\infty}\left[\ln(n) + \gamma + \mathcal{O}\left(\frac{1}{n}\right) - \ln(n)\right]$$

The $\ln(n)$ terms cancel perfectly:

$$C = \lim_{n\to\infty} \left[\gamma + \mathcal{O}\left(\frac{1}{n}\right)\right] = \gamma$$

Final Answer:

$$\gamma$$